



COMPONENTS

Air quality sensor



Renesas Electronics Corporation has enhanced its ZMOD4410 Indoor Air Quality (IAQ) sensor series with the power of embedded artificial intelligence (e-AI), enabling smart odour sensing for ventilation systems, bathroom monitoring and controls, and air quality monitors.

The reinforced platform combines neural network-trained firmware on various microcontroller units (MCUs), such as RL78 MCU, to provide higher resolution measurement results.

With these new capabilities, the ZMOD4410 is not only capable of detecting gases that indicate the presence of certain odours in small enclosed rooms with higher accuracy and improved part-to-part deviation but can also distinguish between sulphur and ethanol-based odours.

Renesas' unique software-configurable ZMOD platform provides greater design flexibility for smart sensing systems, allowing firmware updates with application-specific capabilities, such as selective measurements, to detect volatile organic compounds (VOCs).

Renesas

www.renesas.com

Body temperature MCU

Health monitoring at home and in public places has become crucial these days. Due to the impact of the Covid-19 pandemic, the world has drastically changed in its



operations in public places by checking health conditions through temperature monitoring. In such conditions, rapid measurement and accuracy play an important role.

Keeping this in mind, Holtek has

announced BH67F2742, which is a dedicated MCU for contactless infrared thermometer applications. The MCU flash architecture facilitates production parameter adjustment, product calibration, and data storage. Compared with traditional solutions, the integrated LDO, EEPROM, LED and LCD drivers, low noise operational amplifier, and 24-bit A-D converter reduce the requirements for large numbers of external components.

The BH67F2742 has a built-in LCD driver used to drive an LCD to display temperature. Buzzer and motor are used for vibration and sound output alert when the measurement is complete.

Holtek

www.holtek.com

Car door lock IC

STMicroelectronics has introduced L99UDL01, an automotive universal door-lock integrated circuit (IC)



that integrates six MOSFET half-bridge outputs and two half-bridge gate drivers with protection and diagnostic functions.

As a complete centralised solution for electronically driving car locks from a body control module (BCM), the IC replaces multiple individual motor drivers and associated analogue and passive components while providing sophisticated functionalities.

It comes with unique safety features for normal operation in the event of an accident and ensures quick access to the vehicle.

Also, the L99UDL01 has two power-saving modes—a 50µA standby mode ready for re-activation by an SPI command from BCM and a sleep mode that reduces current to below 15µA. It is packaged in 10mm × 10mm TQFP64L.

STMicroelectronics

www.st.com

DC-DC converter

TRV 1M by TRACO Power is a 1W DC-DC converter series with reinforced isolation of 5000V



AC for medical and industrial applications. The series offers several models with different input voltages (± 10 per cent) between 5 and 24V DC.

With continuous short circuit protection and low leakage current of less than 2µA, this converter series is suited to protect any connected interfaces or applied parts to patients. Featuring semi-regulated outputs, this series provides a good level of regulation without affecting the cost-efficiency.

It is an ideal solution for applications where a completely unregulated DC-DC converter would not meet your regulatory requirements.

The device can operate up to 5000m altitude and is now available in a compact SIP-9 package.

TRACO Power

www.tracopower.com

Diode array

Microchip Technology has launched a new series of Transient Voltage Suppressor (TVS) vertical arrays—MDA3KP



TVS. The offering allows military system circuit protection for ruggedised handheld radios, Global Positioning System (GPS), and communication test equipment. Other device applications include automotive, industrial logging equipment, power supplies, and marine systems.

Microchip's MDA3KP TVS diode array family offers an integrated multi-diode solution and provides fast-reacting avalanche breakdown diode (ABD) that can divert excess current around sensitive components to protect them from electrical overstress.

Its vertically constructed 8-diode, 16-pin surface-mount package allows efficient layout and ease of design, thus facilitating less requirement of printed